

GUL BHATNAGAR

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SKILLS

Programming Languages: Java, Python, C++, HTML, CSS, JavaScript

Web Development: Frontend (HTML, CSS, JS, React.js), Backend (Flask, Node.js + Express), Databases (MongoDB, SQLite, SQL)

Data Science & Machine Learning: Supervised ML (XGBoost, Logistic Regression), Natural Language Processing (NLP, Sentiment Analysis, Hugging Face, VADER), Model Deployment (Streamlit, Flask API)

Visualization & Analytics Tools: Excel, Power BI, Tableau, Matplotlib, Seaborn

Frameworks & Libraries: Flask, Scikit-learn, Pandas, NumPy, Matplotlib, React.js, Express.js, Node.js, Hugging Face Transformers

Tools & Platforms: Git & GitHub, VS Code, Postman, MongoDB Compass, Streamlit Cloud, Jupyter Notebook

Non-Technical Skills: Business Analysis & Requirements Gathering, Stakeholder Management, Change Management, Process Improvement, Data Architecture & Mapping, Technical Communication

PROJECTS

AI-POWERED PERSONALIZED STUDY ASSISTANT

Dec 2024-Jan 2025

- Developed a web-based AI assistant used by **50+ students**, generating study plans customized to user inputs.
- Integrated OpenAI's GPT model to respond to **100+ academic queries** per session via Flask backend.
- Designed a user interface and stored personalized data for **100+ sessions** using SQLite.
- Devised robust file-handling mechanisms and intuitive routing functionality.

FairFare- A SPLITWISE CLONE (MERN Stack)

May 2025

- Built and deployed a full-stack expense-sharing app with **100% responsiveness**, enabling user registration, group creation, and bill splitting.
- Developed **15+ RESTful APIs** using Node.js, Express, and MongoDB.
- Frontend built using React and Axios; deployed both frontend/backend on Render with **zero downtime in testing phase**.
- Handled **CORS issues** and integrated environment variables for secure API access.

AI-Mitra- MULTILINGUAL WOMEN'S HEALTH AND WELLNESS CO-PILOT

June 2025

- Built an AI chatbot offering health guidance to **200+ users** in **5+ Indian languages**, with multilingual NLP.
- Integrated **3 ML models** for symptom classification (PCOS, PMS, anemia) and emotion detection with an accuracy of **85%+**.
- Used GPT-4 to generate personalized health education responses and handle **real-time chats with 90%+ accuracy**.
- Backend handled **concurrent 30+ user sessions** via FastAPI; responsive chat frontend built with React.

CRM NLP DASHBOARD (Streamlit, Hugging Face, Scikit-learn, Python)

August 2025

- Engineered and preprocessed **10,000+ telco customer records** (categorical encoding, scaling, feature selection) to train a **churn prediction model** achieving **82% accuracy, 0.79 precision, and 0.81 recall**.
- Developed an **NLP sentiment analysis pipeline** integrating Hugging Face Transformers and VADER to classify customer feedback with **95% accuracy across 2 sentiment classes**.
- Built and deployed an interactive **CRM dashboard on Streamlit Cloud**, enabling real-time **customer churn probability (0-1) prediction and sentiment insights**, improving decision-making speed by **~40%**.

EXPERIENCE

Full Stack Developer Intern

Jun 2025-Oct 2025

Karvatek Solutions, (Remote)

- Developed **3+ scalable web applications** using the MERN stack (MongoDB, Express, React, Node.js). Collected data from various data sources increasing data volume by **30%**
- Built and integrated **10+ RESTful APIs**, managed **5+ dynamic UI components** using React, and handled backend logic with Node.js.
- Collaborated with a team of **5 developers** on live client projects and followed agile sprints with **bi-weekly deliverables**.
- Gained practical exposure to **version control (Git)**, **deployment pipelines**, and **manual testing of 20+ features**.

Machine Learning Intern

Nov 2024-Jan 2025

smartData Enterprises, (Remote)

- Developed and tested **5 machine learning models** for diverse business use-cases using Python and Scikit-learn.
- Performed data preprocessing and model tuning on **100K+ data rows**, improving accuracy by **12-18%** across iterations.
- Collaborated with **3 data scientists** to analyze real-world datasets and enhance debugging workflows for **5+ projects**.

EDUCATION

Vellore Institute of Technology

Jun 2022 - Jun 2026

Bachelor of Engineering in Computer Science : **GPA: 8.20/10**

Relevant Coursework: Operating Systems, Computer Networks, OOPS, Data Science, Machine Learning, Database Management Systems

CERTIFICATIONS

AWS Cloud Practitioner || AWS

Jan 2024

Machine Learning Specialization || Stanford online

Dec 2024

Google Data Analysis Professional Certificate || COURSERA

Jun 2024

Data Visualisation with Power BI || Great Learning

July 2025

Oracle Cloud Infrastructure Data Science Professional Certificate || Oracle

Aug 2025

Advanced DSA using C++ || IamNeo

May 2024